

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 9.1b: The Agentic-AI Landscape (mid-2026)

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Topic 9.1b Agenda

T1 framed the terminal-agent workflow. This interlude widens the lens to the broader agentic-AI landscape an economist now operates inside.

1. **The interoperability stack** — MCP, A2A, AGNTCY/ACP, neutral governance
2. **Orchestration patterns** — frameworks, context engineering, the multi-agent cost-benefit
3. **Reasoning models & inference-time compute** — think budgets, measured time horizons
4. **Autonomous coding & computer-use agents** — scheduled runtimes, sandboxing
5. **Benchmarks & measurement validity** — ties to the course's measurement theme
6. **Safety & reproducibility** — untrusted inputs, chain-of-custody

Reading convention: [solid] = primary or citable; [hype] = vendor self-report (counter-source given).

Facts current as of June 2026. Model versions are stated with release dates; the field moves monthly — recheck before teaching.

The Interoperability Stack

Compete on models, cooperate on plumbing

Three Layers of Agent Interoperability

The 2025–26 story is standard-setting: vendors compete on models but increasingly cooperate on the wiring between agents and tools.

Layer	Protocol	What it standardizes
Agent ↔ tool	MCP (Model Context Protocol)	how an agent discovers and calls external tools/data sources
Agent ↔ agent	A2A (Agent-to-Agent)	how independent agents delegate tasks to one another
Discovery / identity	AGNTCY / ACP	agent directories, capability cards, identity and trust

- MCP was introduced by Anthropic (Nov 2024); it is now *multi-vendor*, not proprietary.
- **OpenAI adopted MCP** at DevDay (Oct 6, 2025); Google and Microsoft tooling support it too. **[solid]**
- Governance moved to a neutral home: the **Linux Foundation Agentic AI Foundation (AAIF)**, formed Dec 9, 2025, now hosts MCP and related projects. **[solid]**

Why Neutral Governance Is a Network-Effects Story

For economists, the interoperability stack is a textbook standard-setting / network-effects case.

- A shared protocol lowers switching costs: a tool wrapped once in MCP works across Claude, ChatGPT, Gemini, and open agents
- Network effects make the standard more valuable as adoption grows — which is why a single-vendor protocol faces adoption resistance
- Moving MCP under the **Linux Foundation AAIF** (Dec 9, 2025) is the classic move: hand the “plumbing” to a neutral body so rivals will build on it
- Competition then concentrates where rents remain: model quality, latency, and price — not the connectors

Reproducibility note. Protocols version. Pin the *spec version* you used (e.g. MCP spec date, A2A version) alongside the model version and date — a tool call that worked under one spec revision may behave differently under the next.

Orchestration Patterns

How multi-step agent systems are actually built

Four Orchestration Patterns

Production agent systems are built from a small set of recurring control structures.

Pattern	Shape
Orchestrator–worker	a lead agent decomposes a task and dispatches workers; <i>dominant in production</i>
Planner–executor	one component plans, another executes step by step
Reflection	the agent critiques and revises its own output before returning
Handoff	control passes between specialized agents (e.g. triage → specialist)

The terminal-agent workflow you learned in T1–T5 is an orchestrator–worker system: the main agent orchestrates; code-reviewer, domain-reviewer, and verifier sub-agents are the workers. The landscape just adds vocabulary and tooling around the same loop.

Frameworks Reached 1.0 (2025–26)

The orchestration layer matured from research demos to versioned, supported frameworks.

Framework	Milestone	Vendor
LangGraph 1.0	Oct 2025	LangChain
OpenAI Agents SDK (+ AgentKit)	SDK Mar 2025	OpenAI
Google Agent Dev. Kit (ADK)	v1.0 May 2025	Google
Microsoft Agent Framework	1.0 Apr 2026	Microsoft
CrewAI	1.0	CrewAI
Anthropic Claude Agent SDK	Sep 2025	Anthropic

Naming note. What was the *Claude Code SDK* was renamed the **Claude Agent SDK** on Sep 29, 2025, reflecting generalization beyond coding. **[solid]**

Durable lesson: the patterns outlive the framework. Learn orchestrator–worker once; map it onto whichever SDK your institution adopts.

The Multi-Agent Cost–Benefit Decision

More agents is not free. Treat it as marginal value vs. marginal cost.

- Anthropic reports its multi-agent research system beat a single-agent baseline by **+90.2%** on its internal research-eval. . .
- . . . at roughly **15× the token cost**
- So multi-agent pays off when the task is broad, parallelizable, and the answer's value is high relative to tokens
- It is wasteful for narrow, sequential, or cheap tasks

[solid, label: **Anthropic-internal-eval**] — a vendor benchmark on a self-defined research task, not an external suite. Read the +90.2% as directional, not portable.

Economist framing. You are choosing a point where marginal research value = marginal token (and latency, and debugging) cost. The Aiyagari sub-agent review in T4 was justified on *wall-clock*, not because “more agents are better.”

Context Engineering Displaced Prompt Engineering

The 2025–26 shift: managing what is in the context window now matters more than wording a clever prompt.

- **Compaction.** Summarize and clear stale turns so the working set stays in the high-recall part of the window (the “lost in the middle” problem from Lec07 T3b).
- **External memory.** Persist state to files (`NOTES.md`, `progress.md`, `CLAUDE.md`) rather than relying on the chat history.
- **Sub-agents with clean context windows.** Hand a worker a fresh, focused context instead of the orchestrator’s accumulated noise.

Why this matters for research: it is exactly the discipline T4–T5 already taught — file-driven workflow, progress logs, focused reviewers. The landscape gave it a name; the habit was already correct.

Reasoning Models

Tool use in the reasoning loop, and how long agents actually run

Reasoning Models and Think-Budget Control

Frontier models now spend variable inference-time compute, and increasingly call tools inside the reasoning loop.

- **Tool use in the loop.** OpenAI o3 / o4-mini (Apr 2025) interleave tool calls with chain-of-thought rather than reasoning first and acting after.
- **Think-budget routers.** GPT-5 (Aug 2025) → GPT-5.5 (Apr 2026) route between fast and deep modes per query.
- **Toggleable extended thinking + configurable token budget.** **Claude Opus 4.5** (Nov 2025) lets you turn extended thinking on/off and cap the thinking-token budget; it was the first model past **80% on SWE-bench Verified**, and Opus pricing was cut to \$5/\$25 per Mtok (input/output).
- **Parallel deep reasoning.** Gemini “Deep Think” explores multiple reasoning paths in parallel.

Benchmark caveat (carry this everywhere). The 80% SWE-bench Verified figure is real but the metric is contaminated/saturating: OpenAI *stopped reporting* SWE-bench Verified in Feb 2026. Pair any Verified headline with SWE-bench Pro (~55–59% at the frontier, held-out). **[solid]**

How Long Can an Agent Actually Work? Measure, Don't Trust Marketing

Vendors say “works for hours” or “runs for days.” METR gives a measured alternative.

- METR's *task-length-doubling* result: the task length an agent completes at **50% reliability** has doubled roughly every **7 months**, accelerating to **~4 months** in 2024–25. **[solid]**
- Time Horizon 1.1 (Jan 2026) pegs **Opus 4.5 $\approx$ 320 min** and **GPT-5 $\approx$ 214 min** (50%-reliability horizon).
- Use these measured minutes instead of vendor “multi-hour/multi-day autonomy.”

Metric fragility. The number is tied to a **50% reliability** threshold and a specific task suite; the horizon at 80% reliability is much shorter, and it shifts across suites. A measured-but-fragile number still beats a vendor claim.

[hype counterpoint] product pages advertising “multi-hour” or “multi-day” autonomy — no reliability threshold stated.

Research translation: the right question is not “can it run for hours?” but “at what reliability, on what task distribution, validated how?”

Autonomous Coding & Computer Use

From interactive assistants to scheduled, sandboxed runtimes

Coding Agents Became Scheduled, Autonomous Runtimes

In 2025–26, coding agents shifted from interactive assistants to background runtimes with persistent state.

- **OpenAI Codex “Goal Mode”** became the default (May 2026), with persistent cross-session state — the agent resumes a goal rather than starting fresh each chat.
- **Anthropic “Code with Claude” (May 6, 2026)** reframed Claude Code as an *orchestrator*:
 - **Auto Mode** with prompt-injection / destructive-action classifiers + human approval gates
 - **Routines** triggered by cron or webhook
 - **Worktrees** for parallel branches
 - sandboxed **Managed Agents** with scoped credentials

Lesson for research infra. Before an agent runs unattended on real data or repos, configure sandboxing, credential scoping, and approval gates first. “Autonomous” is a permissions decision, not a model capability.

Computer-Use Agents: Strong Benchmarks, Volatile Products

Browser/computer-use agents got good on paper but the standalone products churned.

- Computer-use / browser agents reached **~78–84% on OSWorld-Verified** (human baseline ~72%).
- **But standalone products are volatile:** Google *killed* Project Mariner (May 2026); OpenAI folded Operator into ChatGPT Agent / Atlas.

Flag the benchmark numbers. OSWorld scores are **scaffold-dependent and largely self-reported**: the same model scores very differently under different harnesses, and “above human baseline” does not mean reliable on *your* unseen workflow. Treat as directional. **[solid claim that products churned; benchmark figures = vendor/scaffold-dependent]**

Lesson: do not build durable research infrastructure on a single experimental product. Build on the protocol layer (MCP) and your own files; treat any one browser agent as replaceable.

Benchmarks & Measurement Validity

The course's measurement theme, applied to agents

SWE-bench Verified Collapsed Under Scrutiny

This module is the measurement theme of the course pointed at agent evals: a headline number is not a measurement until you know its validity.

- Contamination + harness-hacking critiques eroded SWE-bench Verified; OpenAI stopped reporting it in Feb 2026.
- Berkeley RDI *trivially broke four agent benchmarks via the harness* (Apr 2026) — exploiting the scaffold, not solving the task.
- **Always pair** a Verified headline with **Scale's SWE-bench Pro** (~55–59% frontier, held-out problems).
- **Reliability under repetition** matters more than best-of- n for autonomous use: τ -bench *pass-hat- k* falls from ~90% single-try to ~57% at $k=8$.

Hard rule. Never put a SWE-bench *Verified* score and a SWE-bench *Pro* score on the same axis — different problem sets, different difficulty, different contamination exposure. **[solid]**

Single-Try Score vs. Reliability Under Repetition

For an agent that runs unattended, the relevant statistic is not its best run — it is whether it succeeds every time.

Best-of- n thinking (wrong for autonomy)

- report the best of several attempts
- flatters the model
- fine for a human-in-the-loop draft you will check

pass-hat- k / reliability (right for autonomy)

- probability *all* k independent runs succeed
- τ -bench: $\sim 90\%$ at $k=1 \rightarrow \sim 57\%$ at $k=8$
- this is what matters when no human checks each run

Economist translation: an unattended agent on a recurring data pull is a repeated trial. A 90% single-try success compounds into frequent failure across a quarter of nightly runs. Design for the reliability curve, not the demo.

Safety & Reproducibility

Untrusted inputs and chain-of-custody for agentic research

Treat Everything the Agent Reads as Untrusted

The defining agentic-AI security failure is prompt injection: data the agent merely reads can hijack what it does.

- **OWASP Top-10 for Agentic Applications** (Dec 2025) now codifies these risks.
- Documented exploits: **EchoLeak** (CVE-2025-32711, zero-click M365 Copilot data exfiltration); **ShadowPrompt** in Claude's Chrome extension; Anthropic's **GTG-1002** — the first largely-autonomous cyberattack (Nov 2025).
- **NIST CAISI** red-teaming raised task-hijacking success from **11% to 81%** with adaptive attacks.

Operating rule. Treat any content the agent reads — emails, web pages, retrieved documents, even tool descriptions — as an *untrusted injection channel*. Use least-privilege sandboxing and revocable permissions, exactly as T4's `settings.json` deny list and T5's security frame prescribed. **[solid]**

Reproducibility: Chain-of-Custody for Agentic Results

Can an agent reproduce a published computational result? Often only partly — which makes logging non-negotiable.

- **CORE-Bench:** agents reproduce published computational results only **~60% (easy)** / **~21% (hard)**. **[solid]**
- Therefore **log full trajectories:** prompts, tool calls, seeds, *model versions*, *spec versions*, and *dates* — the chain-of-custody for any agent-produced artifact.

Read vendor safety numbers skeptically. Distinguish a vendor's “~1% attack success on a *fixed* suite” from academic findings that *adaptive* attacks still exceed ~85%. A fixed-suite number measures the suite, not the threat model. **[solid]**

Ties to T6. This is the same six-criteria discipline (reproducibility, verifiability, auditability) — now with the dates and spec versions the landscape forces you to pin.

Synthesis: The Landscape Reinforces the Workflow Spine

Every module above maps back onto the terminal-agent workflow of T1–T6 and onto the course's measurement / identification theme.

Landscape module	Connection to the spine
Interoperability (MCP/A2A)	build on the neutral protocol layer, not one product; pin spec versions
Orchestration	T4's sub-agent review <i>is</i> orchestrator–worker; pay for multi-agent only when value > token cost
Reasoning & time horizons	“runs for hours” → a measured, reliability-thresholded number
Coding / computer-use	autonomy is a sandboxing + approval-gate decision, set before the run
Benchmarks	a headline score is not a measurement until you know its validity (Verified vs. Pro)
Safety & reproducibility	untrusted-input discipline + logged chain-of-custody (dates, versions, seeds)

The plumbing changes monthly; the research discipline does not.